1. USERS (Authentication Base)

```
users
- id (PK)
- name
- email (UNIQUE)
- password_hash
- role (admin, student, teacher)
- phone
- created_at
- updated_at
```

👨‍🎓 2. STUDENTS

```
students
- id (PK)
- user_id (FK -> users.id)
- date_of_birth
- parent_name
- address
- created_at
```

👩‍🏫 3. TEACHERS

```
teachers
- id (PK)
- user_id (FK -> users.id)
- qualifications
- bio
- availability (JSON)
- created_at
```

📚 4. COURSES

```
courses
- id (PK)
```

```
- title
- description
- category
- fee
- capacity
- created_by (FK -> users.id)
- created_at
```

📚 5. UNITS (Subjects inside courses)

```
units
- id (PK)
- course_id (FK -> courses.id)
- unit_code (e.g. MATH101)
- unit_name
- description
- created_at
```

📅 6. COURSE SCHEDULES

```
course_schedules
- id (PK)
- course_id (FK -> courses.id)
- teacher_id (FK -> teachers.id)
- day_of_week
- start_time
- end_time
- location
```

📝 7. ENROLLMENTS

```
enrollments
- id (PK)
- student_id (FK -> students.id)
- course_id (FK -> courses.id)
```

```
- status (enrolled, pending, cancelled)
- enrolled_at
```

👯 8. SIBLING GROUPS

```
sibling_groups
- id (PK)
- group_name
```

👯 9. SIBLINGS (Mapping)

```
siblings
- id (PK)
- student_id (FK -> students.id)
- sibling_group_id (FK -> sibling_groups.id)
```

🏷️ 10. DISCOUNTS

```
discounts
- id (PK)
- name
- type (percentage, fixed)
- value
- condition (e.g. "siblings >= 2")
- active (boolean)
```

💳 11. PAYMENTS

```
payments
- id (PK)
- student_id (FK -> students.id)
- enrollment_id (FK -> enrollments.id)
- amount
- discount_applied
- final_amount
```

```
- status (paid, pending, failed)
- payment_date
```

🔔 12. NOTIFICATIONS

```
notifications
- id (PK)
- user_id (FK -> users.id)
- message
- is_read (boolean)
- created_at
```

📊 13. ATTENDANCE (Optional)

```
attendance
- id (PK)
- student_id (FK -> students.id)
- course_id (FK -> courses.id)
- unit_id (FK -> units.id)
- date
- status (present, absent)
```

🎓 14. GRADE CODES

```
grade_codes
- id (PK)
- code (HD, D, C, P, F, RPL, WD, DEF)
- name
- description
- is_pass (boolean)
- affects_gpa (boolean)
```

📈 15. GRADE MASTER (Mark Ranges)

```
grade_master
- id (PK)
- grade_code_id (FK -> grade_codes.id)
- min_score
- max_score
- gpa_value
```

📝 16. STUDENT GRADES

```
student_grades
- id (PK)
- student_id (FK -> students.id)
- course_id (FK -> courses.id)
- unit_id (FK -> units.id)
- enrollment_id (FK -> enrollments.id)
- score
- grade_code_id (FK -> grade_codes.id)
- graded_by (FK -> teachers.id)
- graded_at
```

🔗 RELATIONSHIP SUMMARY (IMPORTANT)

```
users → students (1:1)
users → teachers (1:1)

courses → units (1:N)
courses → schedules (1:N)

students → enrollments (1:N)
courses → enrollments (1:N)

students → sibling_groups (M:N via siblings)

enrollments → payments (1:N)

students → student_grades (1:N)
```

units → student_grades (1:N)

grade_codes → grade_master (1:N)
grade_codes → student_grades (1:N)